随机过程

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Review

Bernoulli 试验是独立重复地做一个试验，观察某特定事件 A 是否发生。A 发生（成功）时，记 1，否则记 0. 这个试验结果序列称之为 Bernoulli 过程，其描述涉及以下几个概念：

- 两点分布: $P\{Z=1\}=p$ (成功概率), $P\{Z=0\}=1-p$。独立同分布 （IID）的 01 序列构成 Bernoulli 过程；
- 二项分布 （计数过程）: $(0,n]$ 时间区间内，成功的次数服从二项分布。

$$P\{N(n)=m\}=\binom{n}{m}p^m(1-p)^{n-m}, 0\le m\le n$$

Review

- 几何分布 （间隔过程）: 任给一个时刻，从该时刻起到下次成功发生需要等待的时间服从几何分布。

$$P\{X = k\} = p(1-p)^{k-1}, k \geq 1$$

- 帕斯卡分布: 任给一个时刻，从该时刻起到第 r 次成功发生需要等待的时间服从帕斯卡分布，这个是 r 个几何分布的和。

Review

考虑以下问题：

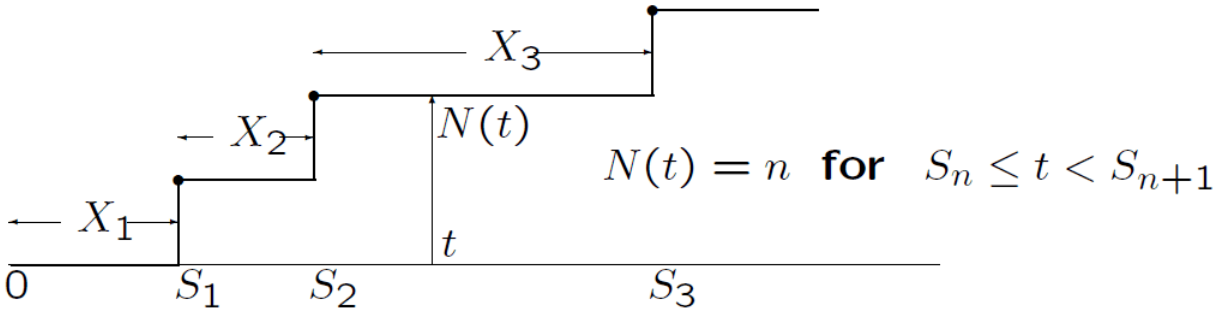
1. 讨论以上各个随机变量的分布律、均值、方差、矩生成函数等。
2. 说明 Bernoulli 分布的无记忆性。
3. 给定计数条件下，成功发生的位置的分布律。
4. Bernoulli 过程的极限过程。

思考题：

1. Bernoulli 过程的样本空间是什么？
2. Bernoulli 过程的树表示。

Arrival Processes

An arrival process is a sequence of increasing rv.s $0 < S_1 < S_2 < \ldots$ where $S_i < S_{i+1}$ means that $X_1 = S_1$ and $X_i = S_i - S_{i-1}$ for $i > 1$ are positive rv.s, $F_{X_i}(0) = 0$. The differences X_i are call interarrival times and the S_i are called arrival epochs.

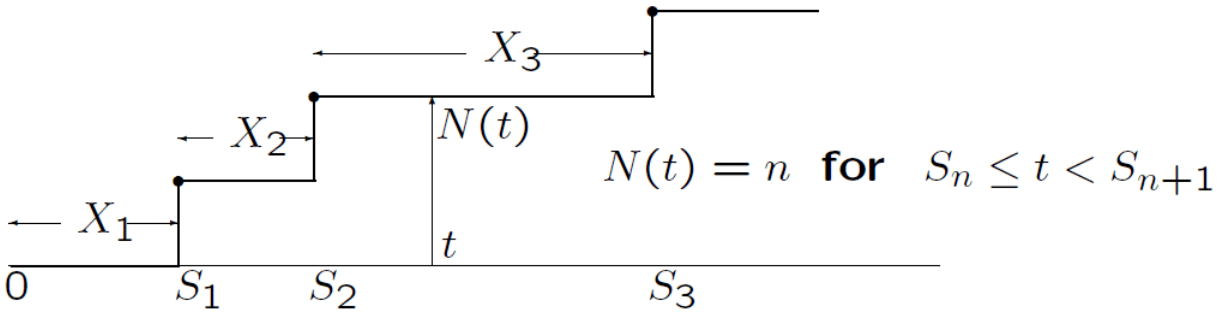


For each $t > 0$, $N(t)$ is the number of arrivals in $(0, t]$. We call $\{N(t) : \ t > 0\}$ an arrival counting process.

Arrival Processes

Remark.

The process starts at time 0 and that multiple arrivals cannot occur simultaneously, or we will permit simultaneous arrivals or arrivals at time 0 as events of zero probability, but these can be ignored.



The figure shows how the arrival epochs, inter-arrival times, and counting variables are interrelated for a generic stair case function.

Arrival Processes

Remark.

- $X_i = S_i - S_{i-1}$ for $i \geqslant 2$ and $X_1 = S_1$; $S_n = \sum_{i=1}^{n} X_i$.
- $\{S_n \leqslant t\} = \{N(t) \geqslant n\}$ for all $n \geqslant 1,\ t > 0$.

If $S_n = \tau$ for some $\tau \leqslant t$, then $N(\tau) = n$ and $N(t) \geqslant n$.

An arrival process can be specified by the joint distributions of the arrival epochs, or of the inter-arrival times, or of the counting random variables.

Definition and properties of a Poission process

Definition 1

A renewal process is an arrival process for which the sequence of interarrival times $X_1, X_2, \ldots$ are i.i.d..

Consider an arrival process $X_1, X_2, \cdots$ where the i-th interarrival time X_i is determined by the outcome of flipping a coin. $\Pr\{X_i = x\} = \frac{1}{2}, x \in \{1, 2\}$. This is a renewal process.

Definition 2

A Poisson process is a renewal process for which each X_i has an exponential distribution,

$$P\{X > x\} = F_X^c(x) = \exp(-\lambda x) \quad \text{for } x \geqslant 0,$$

where λ is a fixed parameter called the rate, or each X_i has the PDF:

$$f_X(x) = \lambda \exp(-\lambda x) \quad \text{for } x \geqslant 0.$$

Memoryless Property

The remarkable simplicity of Poisson processes is closely related to the memoryless property:

Definition 3

A rv. X is memoryless if X is positive and, for all real $t > 0$ and $x > 0$,

$$P\{X > t + x\} = P\{X > t\}P\{X > x\}.$$

Since the interarrival interval for a Poisson process is exponential, i.e., $P\{X > x\} = \exp(-\lambda x)$ for $x \geqslant 0$,

$$\exp(-\lambda(x + t)) = \exp(-\lambda t)\exp(-\lambda x).$$

An arbitrary rv. X is memoryless iff it is exponential.

Memoryless Property

The reason for the word "memoryless" is more apparent when using conditional probabilities,

$$P\{X > t + x | X > t\} = P\{X > x\}.$$

If people in a checkout line have exponential service, and you have waited 15 minute for the person in front, what is his or her remaining service time?

Same as when service started. The remaining waiting time has no 'memory' of previous waiting.
Has your time waiting been wasted?
Why do you move to another line if someone takes a long time?

Memoryless Property

Theorem 1

For a Poisson process of rate λ, and any given $t > 0$, the length of the interval from t until the first arrival after t is a positive r.v. Z with the CDF

$$F_Z(z) = 1 - \exp(-\lambda z) \quad \text{for } z \geqslant 0.$$

This r.v. is independent of both $N(t)$ and the $N(t)$ arrival epochs before time t. It is also independent of the set of r.v.s $\{N(\tau) : \tau \leqslant t\}$.

Idea of proof: Conditional on $N(t) = n$ and $S_n = \tau$, i.e., the number n of arrivals in $(0, t]$ and the time, τ of the most recent arrival in $(0, t]$.

Memoryless Property

Remark.

- This theorem essentially extends the idea of 'memorylessness' to the entire Poisson process.
  That is, starting at any $t > 0$, the interval Z to the next arrival is also an exponential rv. of rate λ. Z is independent of everything before t.
- Let Z_m be the time from the $(m-1)$th arrival after t to the mth arrival epoch after t. We can see that $Z_1, Z_2, \ldots$ are unconditionally i.i.d and also independent of $\{N(\tau) : \tau \leqslant t\}$ and $S_1, S_2, \ldots, S_{N(t)}$.
- Thus the interarrival process starting at t with first interarrival Z, and continuing with subsequent interarrivals is also a Poisson process.

Stationary increment property

Definition 4

A counting process $\{N(t) : t > 0\}$ has the stationary increment property if $N(t') - N(t)$ has the same CDF as $N(t' - t)$ for every $t' > t > 0$.

Let

$$\widetilde{N}(t, t') = N(t') - N(t).$$

Stationary increments means that $\widetilde{N}(t, t')$ has the same distribution as $N(t' - t)$.

Thus, the distribution of the number of arrivals in an interval depends on the size of the interval but NOT the starting point.

Independent increment property

Definition 5

A counting process $\{N(t) : t > 0\}$ has the independent increment property if, for every integer $k > 0$ and every k-tuple of times $0 < t_1 < \cdots < t_k$, the k-tuples of rv.s $N(t_1), \widetilde{N}(t_1, t_2), \ldots, \widetilde{N}(t_{k-1}, t_k)$ are statistically independent.

This implies that the number of arrivals in each of a set of non-overlapping intervals are independent r.v.s.

Theorem 2

Poisson processes have both the stationary increment and independent increment properties.

Probability Density of X and joint density of $X_1, \ldots, X_n$

For a Poisson process $X_1, X_2, \cdots$, where X_i are i.i.d. rv.s with the PDF

$$f_X(x) = \lambda \exp(-\lambda x),$$

then the joint density of $X_1, X_2, \ldots, X_n$ is

$$\begin{aligned} f_{X_1,\ldots,X_n}(x_1, \ldots, x_n) &= \lambda^n \exp(-\lambda x_1 - \lambda x_2 - \cdots - \lambda x_n) \\ &= \lambda^n \exp(-\lambda s_n) \end{aligned}$$

where $s_n = \sum_{i=1}^n x_i$.

Probability Density of S_n and joint density of $S_1, \ldots, S_n$

Definition 6

For a Poisson process, let $S_n = \sum_{i=1}^{n} X_i$, where X_i are i.i.d. rv.s with the PDF

$$f_X(x) = \lambda \exp(-\lambda x),$$

then the S_n has the Erlang density ($S_n \sim \Gamma(n, \lambda)$):

$$f_{S_n}(t) = \frac{\lambda^n t^{n-1} \exp(-\lambda t)}{(n-1)!}.$$

Idea of proof: $S_1 = X_1$ and $S_n = S_{n-1} + X_n$, where S_{n-1} and X_n are independent. Use induction,

$$f_{S_n}(t) = \int_0^t f_{S_{n-1}}(x) \cdot f_{X_n}(t-x) \mathrm{d}x$$

Probability Density of S_n and joint density of $S_1, \ldots, S_n$

Theorem 3

Let $S_1, S_2, \ldots, S_n, \ldots$ be the arrival epoches of a Poisson process with the rate λ. Then the joint density of $S_1, S_2, \ldots, S_n$ is

$$f_{S_1,\ldots,S_n}(s_1, \ldots, s_n) = \lambda^n \exp(-\lambda s_n) \quad \text{for } 0 \leqslant s_1 \leqslant s_2 \leqslant \cdots \leqslant s_n.$$

Given that the n-th arrival is at s_n, the other $n-1$ arrivals are uniformly distributed in $(0, s_n)$, subject to the ordering. Integrating, we get the Erlang marginal density.

The probability mass function for $N(t)$

Theorem 4

For a Poisson process of rate λ, and for any $t > 0$, the PMF for $N(t)$, i.e., the number of arrivals in $(0, t\,]$, is given by the Poisson PMF,

$$P_{N(t)}(n) = \frac{(\lambda t)^n \exp(-\lambda t)}{n!}.$$

proof 1: Use $\{N(t) \geqslant n\} = \{S_n \leqslant t\}$, then

$$\sum_{i=n}^{\infty} P_{N(t)}(i) = \int_0^t f_{S_n}(\tau)\mathrm{d}\tau.$$

$$\begin{aligned} P_{N(t)}(n) &= \int_0^t f_{S_n}(\tau) - f_{S_{n+1}}(\tau)\mathrm{d}\tau \\ &= \frac{\lambda^n}{n!}\int_0^t \exp(-\lambda\tau)\mathrm{d}(\tau^n) + \tau^n \mathrm{d}(\exp(-\lambda\tau)) = \frac{(\lambda t)^n \exp(-\lambda t)}{n!}. \end{aligned}$$

The probability mass function for $N(t)$

proof 2: For some vanishingly small δ,

$$P\{t < S_{n+1} < t+\delta\} = \int_t^{t+\delta} f_{S_{n+1}}(\tau)\mathrm{d}\tau = f_{S_{n+1}}(t)(\delta + o(\delta)).$$

The second way to calculate $\Pr\{t < S_{n+1} < t+\delta\}$ is to first observe that more than one arrival in $(t, t+\delta]$ is $o(\delta)$.

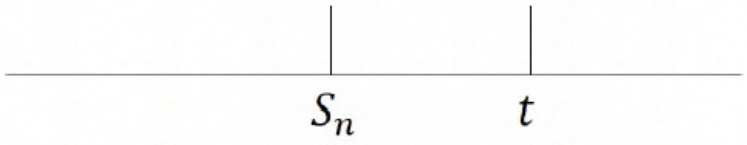
$$\begin{aligned} P\{t < S_{n+1} < t+\delta\} &= P\{t < S_{n+1} < t+\delta | N(t) = n\} P_{N(t)}(n) + o(\delta) \\ &= P_{N(t)}(n)(\lambda\delta + o(\delta)) + o(\delta). \end{aligned}$$

Thus, $P_{N(t)}(n)(\lambda\delta + o(\delta)) + o(\delta) = f_{S_{n+1}}(t)(\delta + o(\delta))$. Dividing by δ and taking the limit $\delta \to 0$, we get

$$\lambda P_{N(t)}(n) = f_{S_{n+1}}(t).$$

注：

$\{N(t)=n\}$意味着$\{S_n \le t\}, \{S_{n+1} > t\}$. 从$S_n \le t$分析，并不方便，因为$S_n$与$t$之间可能还有一段不好处理. 所以，从$\{S_{n+1} > t\}$入手.
用两种方法计算P $\{t < S_{n+1} \le t+\delta\}$，其中$\delta$是一个无穷小量.



方法一：

$$\begin{aligned} \mathrm{P}\ \{t < S_{n+1} \le t+\delta\} &= \int_t^{t+\delta} f_{S_{n+1}}(\tau)d\tau \\ &= f_{S_{n+1}}(t)(\delta + o(\delta)) \end{aligned}$$

这里$f_{S_{n+1}}(t)$ 是Erlang分布，$o(\delta)$表示高阶无穷小.

方法二：

$$\mathrm{P}\ \{t < S_{n+1} \le t+\delta\} = \sum_{k=0}^{n} \Pr\{t < S_{n+1} \le t+\delta, N(t) = k\}.$$

这里把事件分类，$N(t)$可能有情形$k = 0, 1, 2, \ldots, n$.
若$N(t) = k$, 则$(t, t+\delta]$这个区间内必须至少有$n+1-k$次到达发生.
然而, $(t, t+\delta]$中有两个或两个以上到达发生的概率是δ的高阶无穷小. 这是因为, 若到达数目大于等于2，则至少有两个间隔到达时间δ_1, δ_2均小于δ.

该事件发生的概率为$(1-e^{-\lambda\delta_1})(1-e^{-\lambda\delta_2})$. 因此, 我们只需考虑$k = n$的情形.

此时, $(t, t+\delta]$中仅发生一次到达

$$\begin{aligned}\mathrm{P}\left\{t < S_{n+1} \leq t+\delta\right\} &= \mathrm{P}\left\{t < S_{n+1} \leq t+\delta | N(t)=n\right\} \cdot P_{N(t)}(n) + o(\delta) \\ &= (\lambda\delta + o(\delta)) \cdot P_{N(t)}(n) + o(\delta)\end{aligned}$$

通过以上两种方法, 应求得相同结果, 故有

$$P_{N(t)}(n)\lambda = f_{S_{n+1}}(t)$$

$$P_{N(t)}(n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$

Alternate definition of Poisson processes

Definition 2 of Poisson processes

If an arrival process has the stationary and independent increment properties and if $N(t)$ has the Poisson PMF for given λ and all $t > 0$, then the process is Poisson.

Definition 3 of Poisson processes

If an arrival process has the stationary and independent increment properties and satisfies

$$\Pr\{\widetilde{N}(t, t+\delta) = n\} \approx \begin{cases} 1 - \lambda\delta + o(\delta) & \text{for} \quad n = 0, \\ \lambda\delta + o(\delta) & \text{for} \quad n = 1, \\ o(\delta) & \text{for} \quad n \geqslant 2. \end{cases}$$

then it is Poisson.

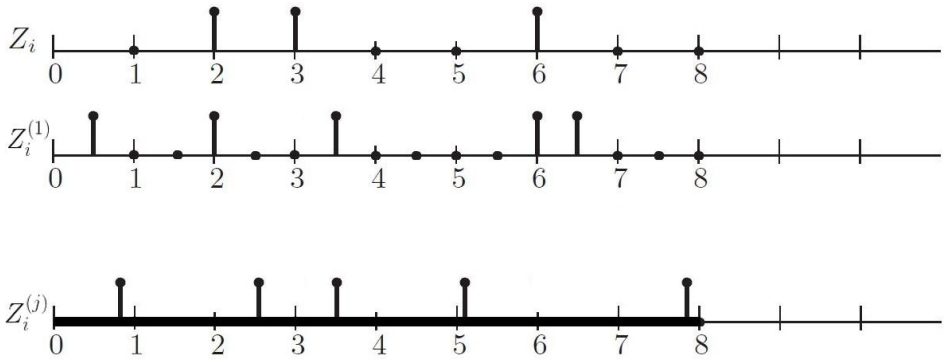
The Poisson process as a limit of shrinking Bernoulli processes

We can view a Bernoulli process as an arrival process, an arrival occurs at discrete time n if and only if $Y_n = 1$. Thus $S_n = Y_1 + \cdots + Y_n$ is the number of arrivals up to and including time n.

$$P_{N(t)}(k) = \binom{\lfloor t \rfloor}{k} p^k (1-p)^{\lfloor t \rfloor - k}, \text{ for } k < \lfloor t \rfloor$$

.

Now we "shrink" the time scale of the process so that for some integer $j > 0$, $Y_i^{(j)}$ is an arrival or no arrival at time $i2^{-j}$.



In order to keep the arrival rate constant, we let $p = \lambda 2^{-j}$ for the j-th process.
The expected number of arrivals per unit time is then λ.

Theorem 7 (Poisson's theorem)

Consider the sequence of shrinking Bernoulli processes with arrival probability $\lambda 2^{-j}$ and time-slot size 2^{-j}. Then for every fixed time $t > 0$ and fixed number of arrivals n, the counting PMF $P_{N_j(t)}(n)$ approaches the Poisson PMF (of the parameter λt) with increasing j,

$$\lim_{j\to\infty} P_{N_j(t)}(n) = \frac{(\lambda t)^n \mathrm{e}^{-\lambda t}}{n!}.$$

Proof.

For each j, the jth Bernoulli process has an associated Bernoulli counting process

$$N_j(t) = \sum_{i=1}^{\lfloor t2^j \rfloor} Y_i^{(j)}$$

$$P_{N_j(t)}(n) = \binom{\lfloor t2^j \rfloor}{n} p^n (1-p)^{\lfloor t2^j \rfloor - n}, \text{ for } n < \lfloor t \rfloor$$

where $p = \lambda 2^{-j}$.
Thus,

$$\begin{aligned}\lim_{j\to\infty} P_{N_j(t)}(n) &= \lim_{j\to\infty} \binom{\lfloor t2^j \rfloor}{n} (\lambda 2^{-j})^n (1-\lambda 2^{-j})^{\lfloor t2^j \rfloor - n} \\ &= \lim_{j\to\infty} \binom{\lfloor t2^j \rfloor}{n} \left(\frac{\lambda 2^{-j}}{1-\lambda 2^{-j}}\right)^n \exp[\lfloor t2^j \rfloor \ln(1-\lambda 2^{-j})]\end{aligned}$$

$$\lim_{j\to\infty} P_{N_j(t)}(n) = \lim_{j\to\infty} \binom{\lfloor t2^j \rfloor}{n} \left(\frac{\lambda 2^{-j}}{1-\lambda 2^{-j}}\right)^n \exp[\lfloor t2^j \rfloor \ln(1-\lambda 2^{-j})]$$

$$(\text{use } \ln(1-\lambda 2^{-j}) = -\lambda 2^{-j} + o(2^{-j}).)$$

$$= \lim_{j\to\infty} \binom{\lfloor t2^j \rfloor}{n} \left(\frac{\lambda 2^{-j}}{1-\lambda 2^{-j}}\right)^n \exp(-\lambda t)$$

$$= \lim_{j\to\infty} \frac{\lfloor t2^j \rfloor \cdot \lfloor t2^j - 1 \rfloor \cdots \lfloor t2^j - n + 1 \rfloor}{n!} \left(\frac{\lambda 2^{-j}}{1-\lambda 2^{-j}}\right)^n \exp(-\lambda t)$$

$$(\text{for } 0 \leqslant i \leqslant n-1,\ \lim_{j\to\infty} \lfloor t2^j - i \rfloor \left(\frac{\lambda 2^{-j}}{1-\lambda 2^{-j}}\right) = \lambda t.)$$

$$= \frac{(\lambda t)^n}{n!} \exp(-\lambda t).$$

Exercises

1. 修理一个机器所需要的时间 T 是均值为 1/2(小时) 的指数随机变量.
   (a) 问修理时间超过 1/2 小时的概率是多少?
   (b) 已知修理持续时间超过 12 小时, 问修理时间至少需要 $12\frac{1}{2}$ 小时的概率是多少?
2. 假设你到达一个单服务线的银行, 你发现有 5 个人在银行中, 一个人在接受服务, 其余 4 个人排队等待, 你加入到队尾. 如果服务时间都是速率为 μ 的指数时间, 问你在银行的平均停留时间是多少?
3. 令 X 是指数随机变量. 不做任何计算说出以下哪一个是正确的. 解释你的答案.
   (a) $\mathrm{E}[X^2|X>1]=\mathrm{E}[(X+1)^2]$;
   (b) $\mathrm{E}[X^2|X>1]=\mathrm{E}[X^2]+1$;
   (c) $\mathrm{E}[X^2|X>1]=(1+\mathrm{E}[X])^2$.

4. 假设光临某商店的顾客数服从参数为 4 人每小时的 Poisson 过程。已知商店早上 9:00 开业，求：
(a) 到 9:30 为止恰好一位顾客光临的概率？
(b) 到 11:30 为止总共到达 5 位顾客的概率？

5. 令 $N = (N(t), t \geq 0)$ 是参数为 2 的 Poisson 过程，求：
(a) $P(N(1) = 2)$
(b) $P(N(1) = 2, N(3) = 6)$
(c) $P(N(1) = 2 | N(3) = 6)$
(d) $P(N(3) = 6 | N(1) = 2)$

6. 令 $N = (N(t), t \geq 0)$ 是参数为 2 的 Poisson 过程，求：
(a) $P(N(1) \leq 2)$
(b) $P(N(1) = 1, N(2) = 3)$
(c) $P(N(1) \geq 2 | N(1) \geq 1)$

7. 令 $N = (N(t), t \geq 0)$ 是参数为 2 的 Poisson 过程，求：

(a) $EN(2)$

(b) $EN(1)^2$

(c) $EN(1)N(2)$

(d) $EN(1)N(2)N(3)$

8.

Exercise 1.39 Let $\{X_i;\ i \geq 1\}$ be IID binary rv s. Let $\Pr\{X_i = 1\} = \delta$, $\Pr\{X_i = 0\} = 1 - \delta$. Let $S_n = X_1 + \cdots + X_n$. Let m be an arbitrary but fixed positive integer. Think! Then evaluate the following and explain your answers:

(a) $\lim_{n\to\infty} \sum_{i:n\delta-m\leq i\leq n\delta+m} \Pr\{S_n = i\}$;

(b) $\lim_{n\to\infty} \sum_{i:0\leq i\leq n\delta+m} \Pr\{S_n = i\}$;

(c) $\lim_{n\to\infty} \sum_{i:n(\delta-1/m)\leq i\leq n(\delta+1/m)} \Pr\{S_n = i\}$.